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 Studierichting: Informatica
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$$\int \frac{dx}{(3x+7)\sqrt{x+2}}$$

$$t = x + 2$$

$$dt = dx$$

$$(x = t - 2)$$

$$\begin{aligned} &= \int \frac{dt}{(3x+7)\sqrt{t}} \\ &= \int \frac{dt}{(3(t-2)+7)t^{\frac{1}{2}}} \\ &= \int \frac{dt}{(3t-6+7)t^{\frac{1}{2}}} \\ &= \int \frac{dt}{(3t+1)t^{\frac{1}{2}}} \\ &= \int \frac{dt}{(3t+1)t^{\frac{1}{2}}} \end{aligned}$$

$$u = t^{\frac{1}{2}}$$

$$u^2 = t$$

$$2udu = dt$$

$$\begin{aligned} &= \int 2u \left(\frac{du}{u(3t+1)} \right) \\ &= \int 2 \left(\frac{du}{3u^2+1} \right) \\ &= \int \frac{2du}{3u^2+1} \\ &= 2 \int \frac{du}{3u^2+1} \end{aligned}$$

$$v = \sqrt{3}u$$

$$dv = \sqrt{3}du$$

$$\frac{dv}{\sqrt{3}} = du$$

$$\begin{aligned}
&= \frac{2}{\sqrt{3}} \int \frac{dv}{v^2 + 1} \\
&= \frac{2}{\sqrt{3}} \operatorname{Bgtan}(v) + c \\
&= \frac{2}{\sqrt{3}} \operatorname{Bgtan}(\sqrt{3}u) + c \\
&= \frac{2}{\sqrt{3}} \operatorname{Bgtan}(\sqrt{3}t) + c \\
&= \frac{2}{\sqrt{3}} \operatorname{Bgtan}(\sqrt{3(x+2)}) + c \\
&= \frac{2}{\sqrt{3}} \operatorname{Bgtan}(\sqrt{3x+6}) + c
\end{aligned}$$

References